

Figure 6: Target selection

Source	Destination	Protocol	Info
vmware_bcc:bd	vmware_bcc:bd:f8	arp	192.168.0.2 to 192.168.0.131

Figure 7: Wireshark analysis

Source	Destination	Protocol	Info
172.31.53.10	172.31.53.1	arp	who has 172.31.53.1? 172.31.53.10
172.31.53.10	172.31.53.1	arp	172.31.53.10 is at 00:0C:29:84:M:3A
172.31.53.10	172.31.53.1	arp	172.31.53.10 is at 00:0C:29:84:M:3A

Figure 8: Ethereal analysis of Cain and Abel's ARP spoofing

```
# chmod 770 arpp
Run the script
# ./arpp
```

This is the manual method to do ARP spoofing. We can perform this with the help of some tools like 'Cain and Abel' and Ettercap.

ARP spoofing with the help of Ettercap

Install Ettercap in Linux and launch it. Press the *Sniff* tab and select *Unified sniffing*. A small window appears for you to select the network card. Select interface card (*eth0*) and press *OK*; the interface will be changed. Start sniffing (press *Ctrl+W*). The next task is scanning, so press the *Hosts* tab and click *Scan* for hosts.

After scanning, click on *Hosts list*, which will give online hosts on the LAN. Select the victim as *Add to Target1* and the gateway as *Add to Target 2* as shown in Figure 6.

Press the *Mitm* tab, and then click *ARP poisoning*. Press the *Plugins* tab and then select *repositon_arp*, in order to send packets continuously.

Analyze the network by using Wireshark, which is running on another host. Figure 7 shows the attacker sending the same MAC address to both IP addresses.

Cain and Abel enables you to send spoofed packets to more than one victim. Figure 8 shows the attack on the LAN. Attacker (172.31.53.10) sends a fake ARP reply packet to the gateway (172.31.53.1).

Prevention

One of the ways of prevention is to enter static MAC address entries so that your computer doesn't broadcast

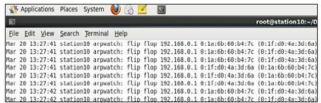


Figure 9: Linux Arpwatch log file

ARP requests. Yet, other routers can still be poisoned and you could be in trouble.

You can add a static entry as follows:

```
>arp -s IP address MAC address
```

Detecting ARP spoofing using open source tools

There is no particular algorithm or tool that can tell you that spoofing is being done. ARP spoofing can be avoided by a static entry in the ARP cache but this is not a good enough solution.

Detection in Linux

ARP spoofing can be detected by some free software. In the Linux environment, Arpwatch can detect ARP spoofing. Arpwatch can be started by using the following command:

```
Service arpwatch restart
```

Arpwatch can be bound with the interface *eth0*, by using the command below:

```
arpwatch -i eth0
```

By default, Arpwatch sends its log details to */var/log/messages*.

To monitor all Arpwatch log messages, use the following command:

```
# tail -f /var/log/messages | grep arpwatch
# cat /var/log/messages | grep arpwatch
```

If ARP spoofing happens, then there is a 'flip flop' in the file as shown in Figure 9.

Detection in Windows

In the Windows environment, the free software DecaffeinatedID acts as an ARP watcher. It maintains a log file, and gives a special alert whenever it sees the MAC address of the gateway change. 

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